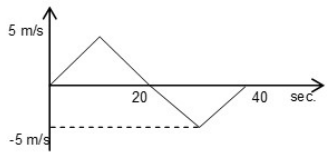
PHYSICS-KM-03

Topics Covered: *Graphs (Extracting Information from given graph + Drawing graph as a problem solving technique)*

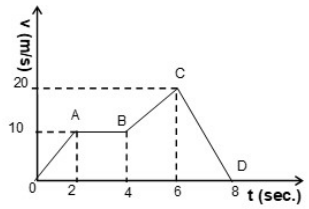
Total Count of Questions: *28*

Expected Time of Attempt by Students: *140 min*

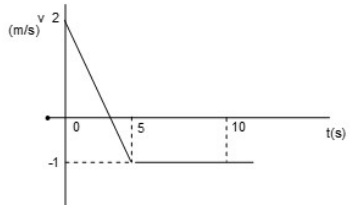
1. From the velocity time graph shown in figure, find the distance travelled by the particle during the first 40 sec. Also find the average velocity during this period.



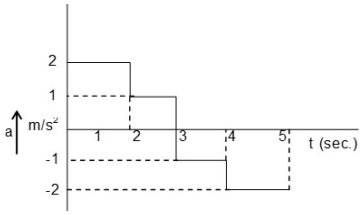
2. Velocity-time graph of a particle moving in a straight line is shown in figure. Plot the corresponding displacement time graph of the particle if at t = 0 displacement is zero.



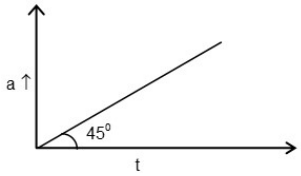
3. The velocity – time graph of a particle moving along X – axis is shown. Find the displacement and the distance travelled in 10 s.



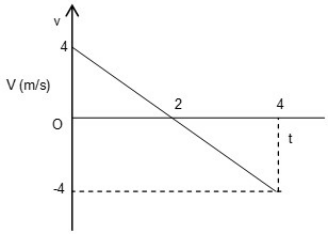
4. A car starting from rest moving on a straight line has acceleration – time graph as shown in the figure. Draw the velocity – time graph.



5. The acceleration of a particle varies with time as shown. If velocity at t = 0 is v_0 then calculate velocity as a function of time.

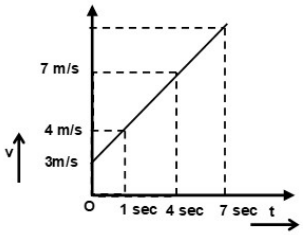


6. A particle starts from origin at t = 0 along +ve x axis. It's velocity –time graph is shown in the figure. Draw
(i) a – t graph (ii) x – t graph

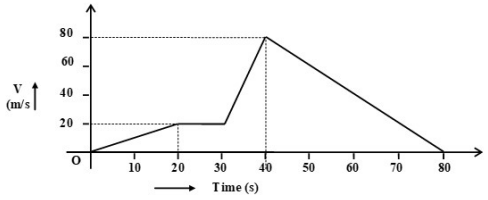


7. A body of mass m is projected vertically upwards with a velocity v_0. It goes up and comes back to the same point. For this motion draw displacement-time, velocity-time, acceleration-time and velocity-displacement graphs.

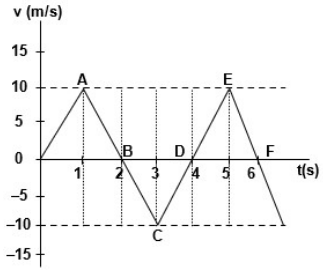
8. The velocity-time graph of a particle is as shown in the figure. Find the distance travelled by the object in 7th second.



9. The velocity time graph of moving object is given in the figure. Draw the acceleration versus time and displacement versus time graph. Find the distance travelled during the time interval when the acceleration is maximum. Assume that the particle starts from origin.

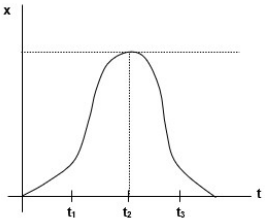


10. From the velocity-time graph of a particle given in figure, describe the motion of the particle qualitatively in the interval 0 to 4 s. Find (a) the distance travelled during first two seconds, (b) the distance travelled during the time 2s to 4s, (c) the distance travelled during the time 0 to 4s, (d) displacement during 0 to 4s, (e) acceleration at t = 1/2 s and (f) acceleration at t = 2s.

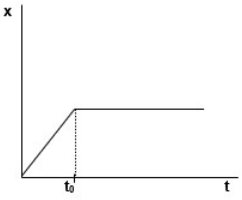


11. A ball is dropped from a height of 20 m above the ground. It rebounds from the ground and raises itself up to the same height. Take the starting point as the origin and vertically downward as the positive X-axis. Draw approximate plots of x versus t, v versus t and a versus t. Neglect the small interval during which the ball was in contact with the ground. (Take g = 10 m/s^2)

12. Figure shows the x coordinate of a particle as a function of time. Find the signs of v_x and a_x at t = t_1, t = t_2 and t = t_3.

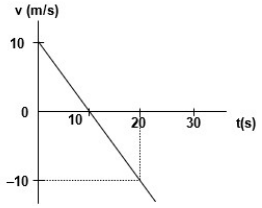


13. Figure shows the displacement time graph of a particle moving on the X-axis. Choose the correct option(s)

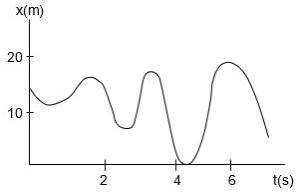


(a) the particle is continuously going in positive x direction
(b) the particle is at rest
(c) the velocity increases up to a time t_0, and then becomes constant
(d) the particle moves at a constant velocity up to a time t_0, and then stops

14. The velocity-time plot for a particle moving on a straight line is shown in the figure. Choose the correct option(s)
(a) The particle has a constant acceleration
(b) The particle has never turned around
(c) The particle has zero displacement
(d) The average speed in the interval 0 to 10 s is the same as the average speed in the interval 10 s to 20 s.

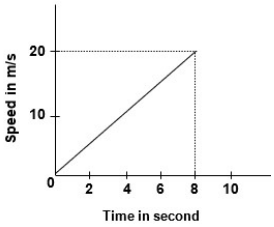


15. Figure shows the position of a particle moving on the X-axis as a function of time

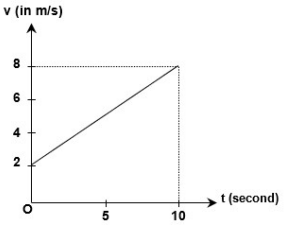


(a) The particle has come to rest 6 times
(b) The maximum speed is at t = 6 s.
(c) The velocity remains positive for t = 0 to t = 6 s
(d) The average velocity for the total period shown is negative

16. The speed of a car as a function of time is shown in figure. Find the distance travelled by the car in 8 seconds and its acceleration.



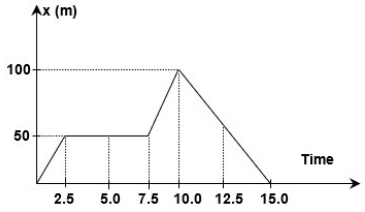
17. Figure shows the graph of velocity versus time for a particle going along the X-axis. Find
(a) the acceleration,
(b) the distance travelled in 0 to 10 s and
(c) the displacement in 0 to 10 s.



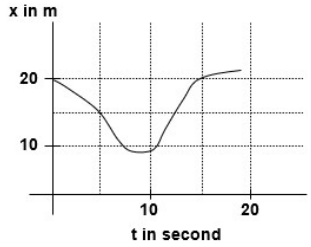
18. Figure shows the graph of the x-coordinate of a particle going along the X-axis as a function of time. Find

(a) the average velocity during 0 to 10s,

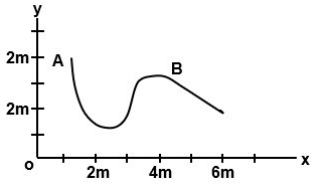
(b) instantaneous velocity at 2, 5, 8 and 12s.



19. Figure shows x-t graph of a particle. Find the time t such that the average velocity of the particle during the period 0 to t is zero.



20. A particle starts from a point A and travels along the solid curve shown in figure. Find the approximately the position b of the particle such that the average velocity between the positions A and B has the same direction as the instantaneous velocity at B.



21. A particle performs rectilinear motion in such a way that its initial velocity has opposite direction with its uniform acceleration. Let x_A and x_B be the magnitude of displacements in the first 10 seconds and the next 10 seconds, then:

(a) $x_A < x_B$

(b) $x_A = x_B$

(c) $x_A > x_B$

(d) the information is insufficient to decide the relation of x_A with x_B.

22. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1, next 10 s is x_2 and the last 10 s is x_3. Then $x_1 : x_2 : x_3$ is the same as

(a) 1 : 2 : 4 (b) 1 : 2 : 5

(c) 1 : 3 : 5 (d) 1 : 3 : 9

23. The displacement time graphs of two particles A and B are straight lines making angles of respectively 30° and 60° with the time axis. If the velocity of A is v_A and that of B is v_B, then the value of $\frac{v_A}{v_B}$ is

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$ (d) $\frac{1}{3}$

24. Two objects moving along the same straight line are leaving point A with an acceleration a, 2 a & velocity 2 u, u respectively at time t = 0. The distance moved by the object with respect to point A when one object overtakes the other is :

(a) $\frac{6u^2}{a}$ (b) $\frac{2u^2}{a}$

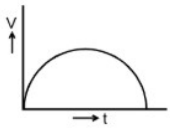
(c) $\frac{4u^2}{a}$ (d) None of these

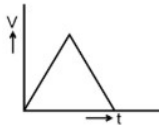
25. A particle is moving along x axis with constant acceleration. At t = 0 , the particle is at $x = 3m$ and $\frac{dx}{dt} = +4m/s$. The maximum value of x co-ordinate of the particle is observed 2 seconds later. Starting from t = 0 sec after what time particle reaches its initial position again?

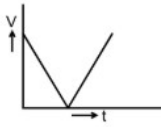
(a) 4 sec (b) 6 sec

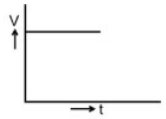
(c) 8 sec (d) 12 sec

26. A particle is thrown vertically upwards. The graph between its speed v and time t is given by (neglecting the air resistance) the following figure :

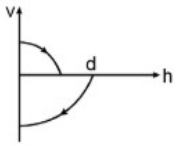
(a) 

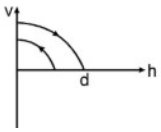
(b) 

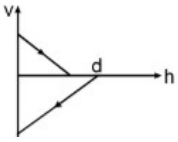
(c) 

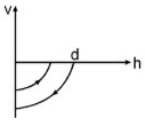
(d) 

27. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height d/2. Neglecting subsequent motion and air resistance, its velocity v varies with the height h above the ground as

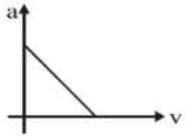
(a) 

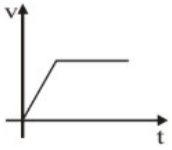
(b) 

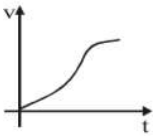
(c) 

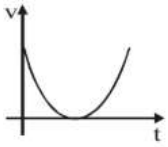
(d) 

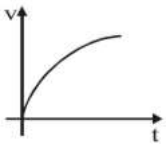
28. Acceleration versus velocity graph of a particle moving in a straight line starting from rest is as shown in figure. The corresponding velocity – time graph would be: -



(a) 

(b) 

(c) 

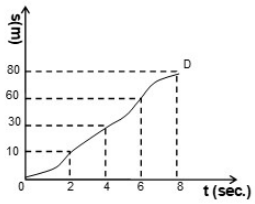
(d) 

ANSWERS

1. Distance = 100 m

Velocity = zero

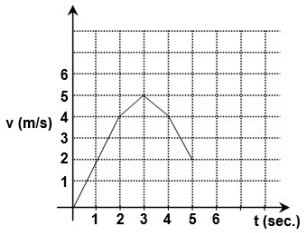
2.



3. Displacement $= \frac{1}{2}\left(\frac{10}{3}\right)(2) - \frac{1}{2}\left(5 - \frac{10}{3}\right)(1) - 5\times 1 = -\left(\frac{15}{6}\right)m$

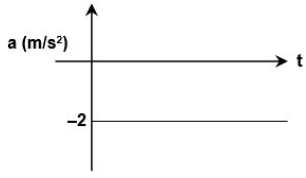
Distance travelled $= \frac{1}{2}\left(\frac{10}{3}\right)(2) + \frac{1}{2}\left(5 - \frac{10}{3}\right)(1) + 5\times 1 = \left(\frac{55}{6}\right)m$

4.

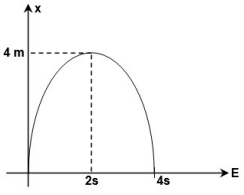


5. $v = v_0 + t^2/2$

6.

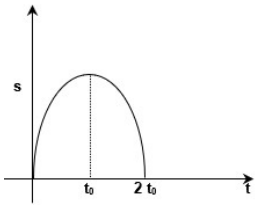


(i)

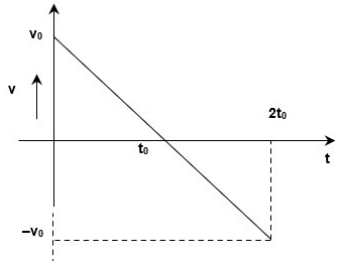


(ii)

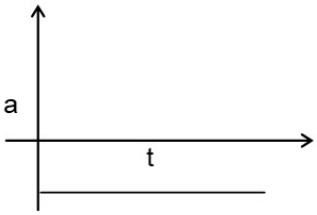
7. Displacement time $t_0 = \frac{v_0}{g}$



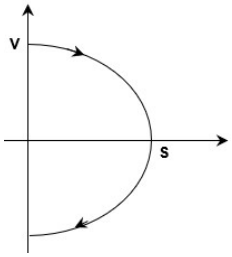
Velocity time



Acceleration time

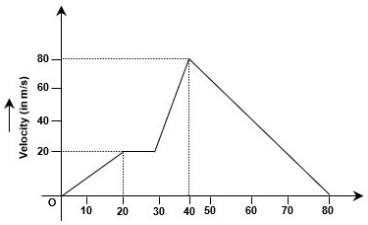


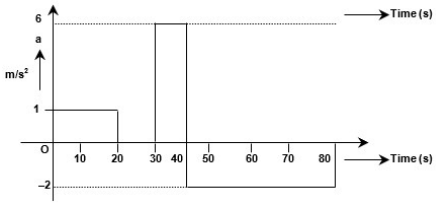
Velocity displacement

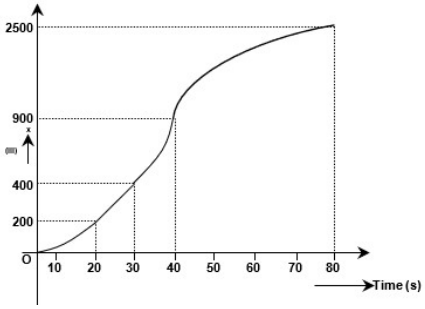


8. 9.5 m

9.







10. (a) 10 m

(b) 10 m (The particle has moved in negative x direction during this period)

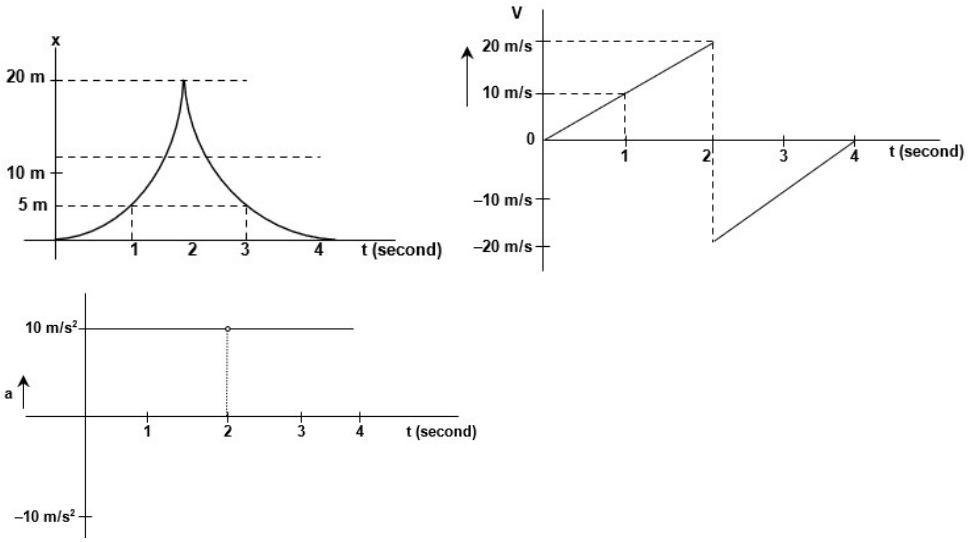
(c) 20 m

(d) 0

(e) 10 m/s^2

(f) –10 m/s^2

11.



12.

Time Instant	Sign of V_x	Sign of a_x
t_1	+	+
t_2	0	–
t_3	–	+

13. (d)

14. (a, d)

15. (a, d)

16. 80 m, 2.5 m/s^2

17. (a) 0.6 m/s^2

(b) 50 m

(c) 50 m

18. (a) 10 m/s

(b) 20 m/s, zero, 20 m/s, –20 m/s

19. 15 s

20. $x_B = 4m$

21. (d)

22. (c)

23. (d)

24. (a)

25. (a)

26. (c)

27. (a)

28. (d)